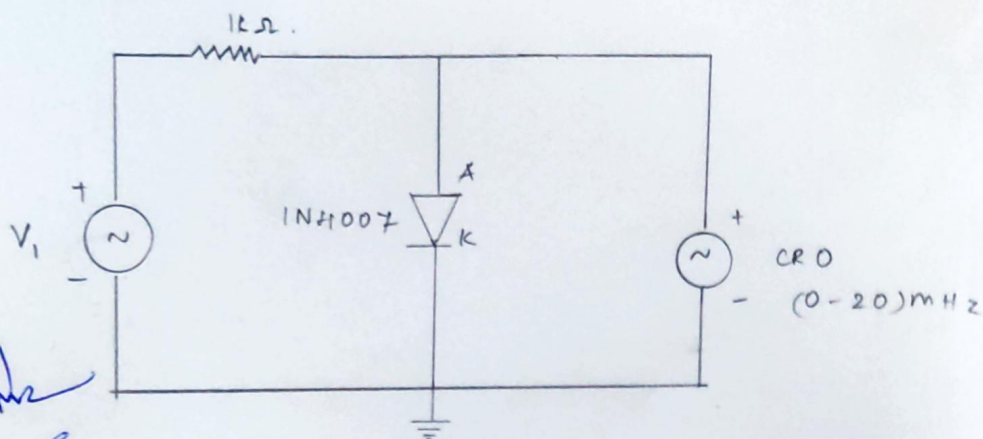
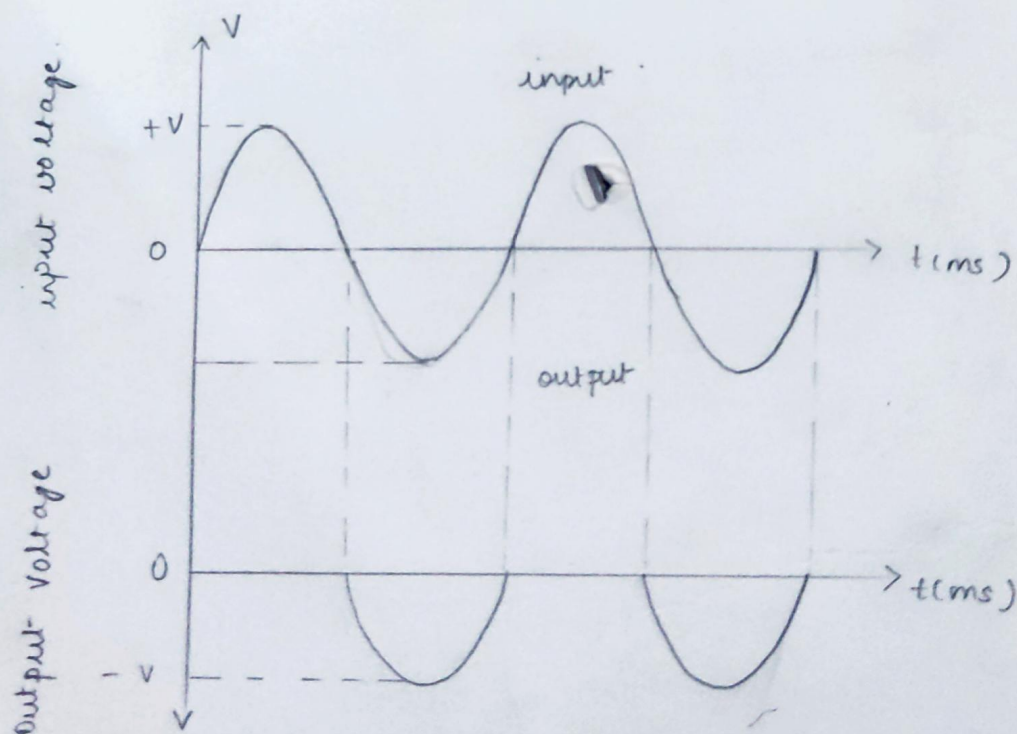


circuit diagram :

positive clipper :



Model graph :-



EX NO: 15a

CLIPPER CIRCUITS USING  
DIODE.

1.08.17

AIM:

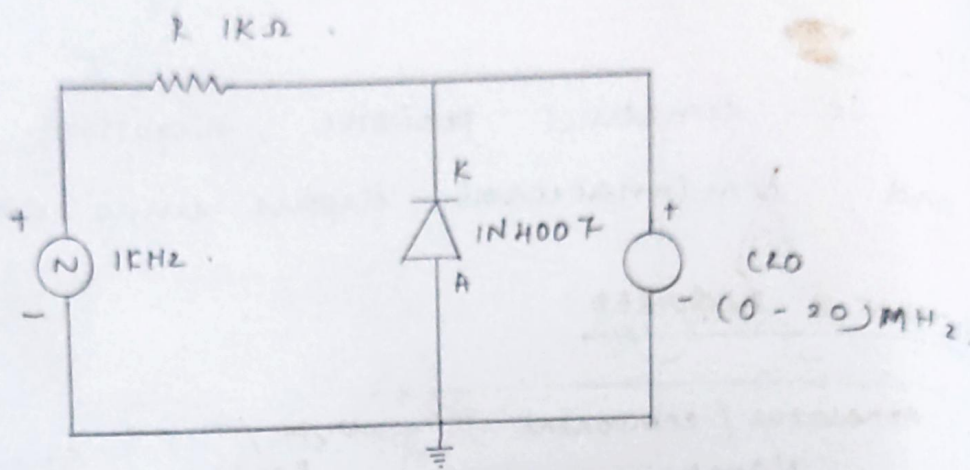
To construct positive, negative, biased and combinational clippers using diodes.

APPARATUS REQUIRED:

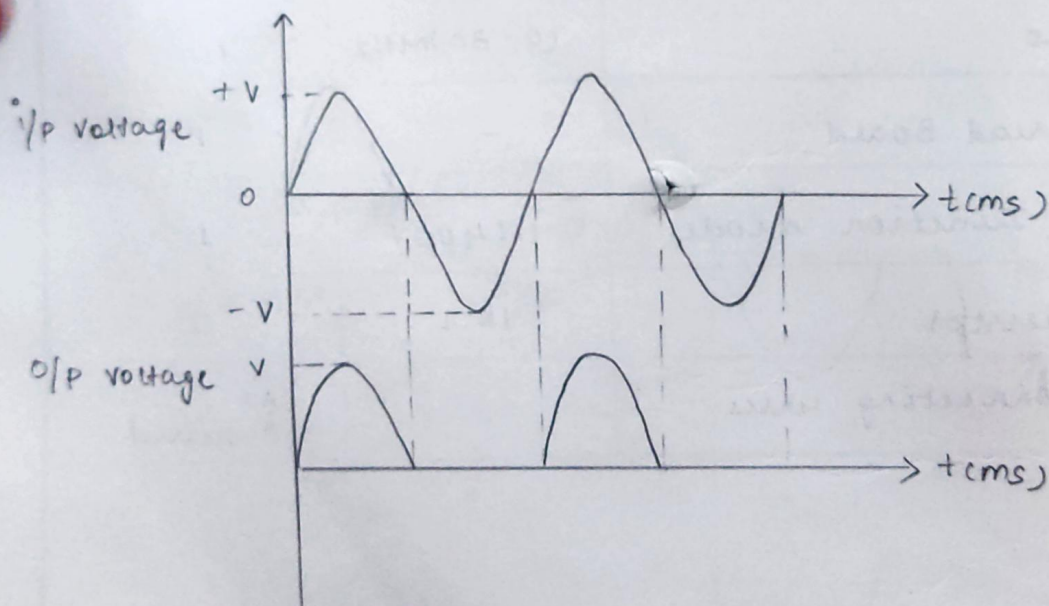
| SL NO. | Apparatus / component Name.       | Type / Range | Quantity.   |
|--------|-----------------------------------|--------------|-------------|
| 1.     | Dual regulated power supply (RPS) | (0-30) V     | 1           |
| 2.     | Function Generator                | (0-30) MHz   | 1           |
| 3.     | CRO                               | (0-30) MHz   | 1           |
| 4.     | Bread Board                       | -            | 1           |
| 5.     | PN Junction diode                 | 1N4007       | 1           |
| 6.     | Resistor                          | 1K $\Omega$  | 2           |
| 7.     | connecting wires                  | -            | As required |



## Negative clipper



### Model graph:

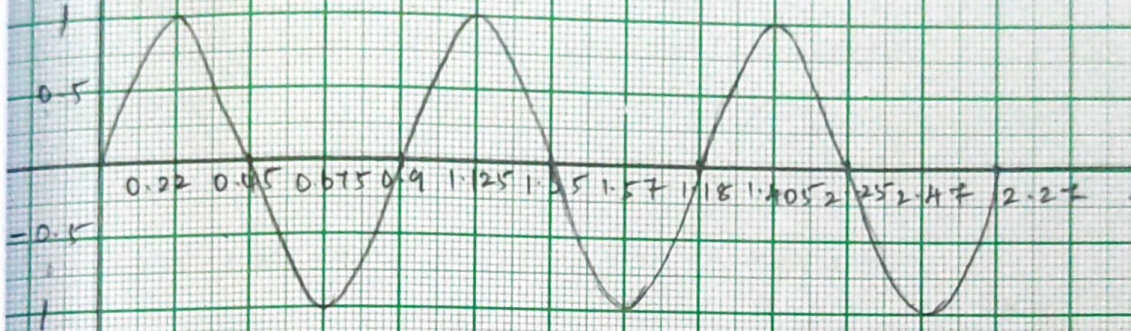


SCALE

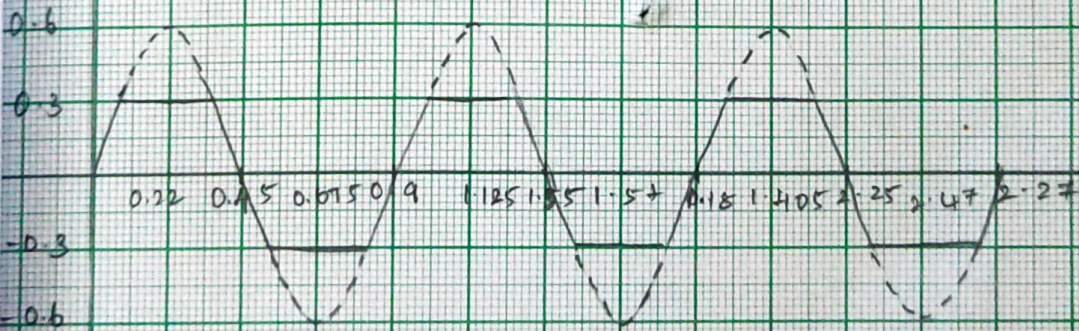
2 cm  $\rightarrow$  1 cm = 0.25 sec

4 cm  $\rightarrow$  1 cm = 0.5 V

INPUT COMBINATIONAL CLIPPER.



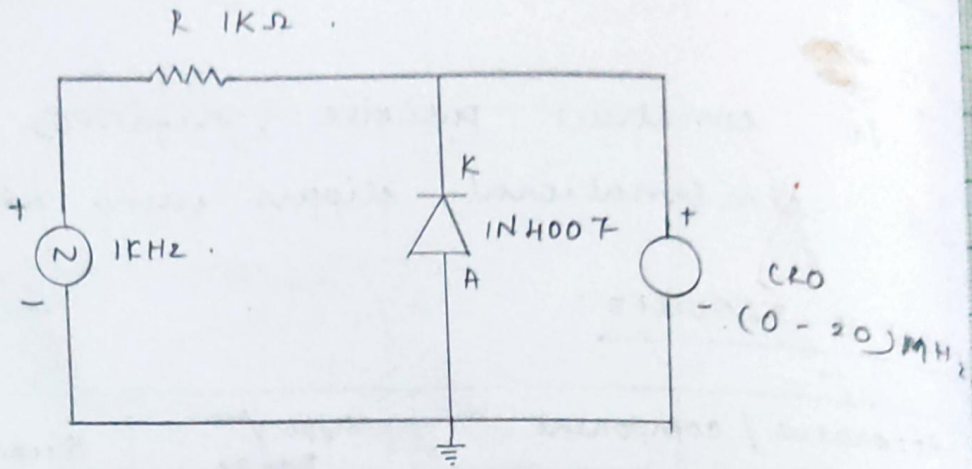
OUTPUT COMBINATIONAL CLIPPER.



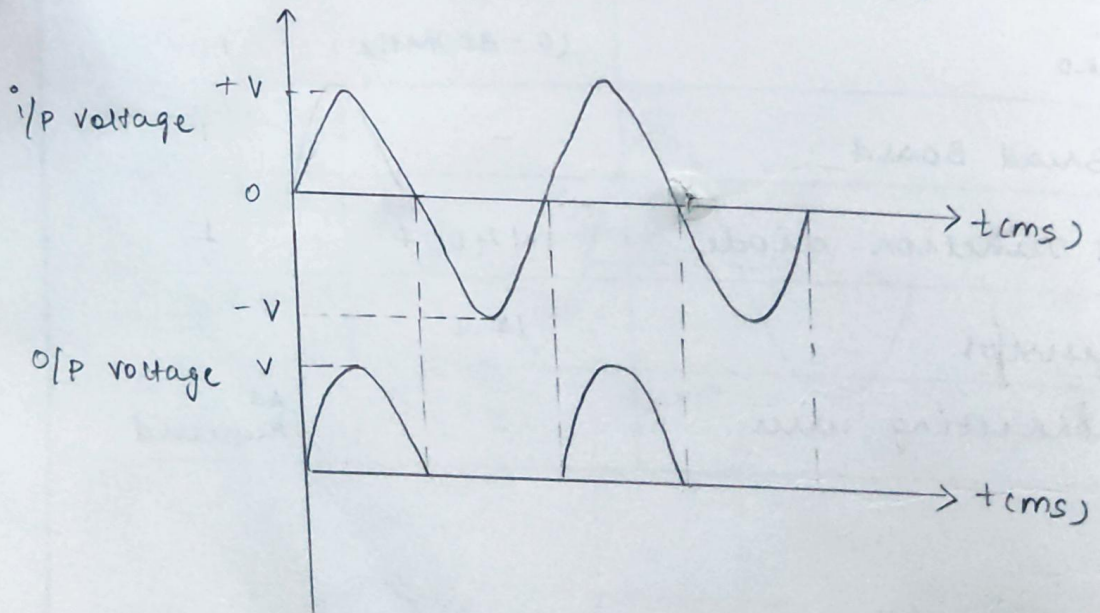
Gain



## Negative clipper:



### Model graph:

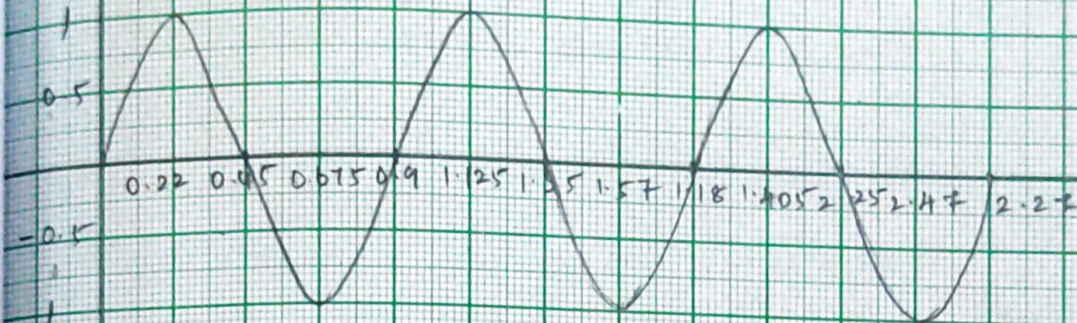


SCALE

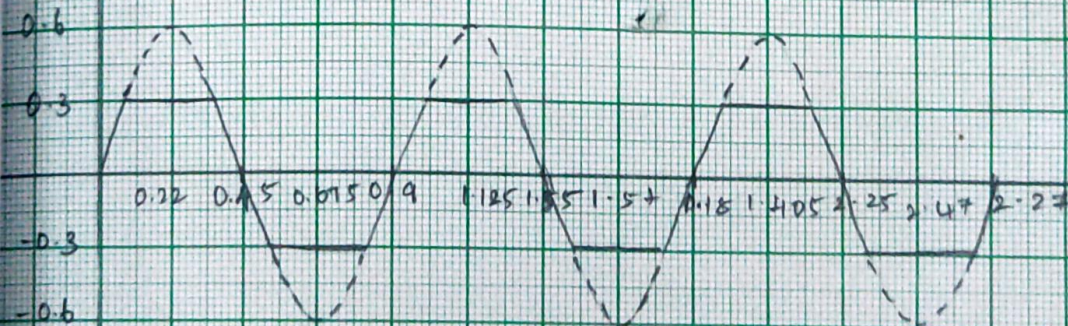
2 cm = 0.25 sec

4 cm = 1.5 V

INPUT COMBINATIONAL CLIPPER



OUTPUT COMBINATIONAL CLIPPER



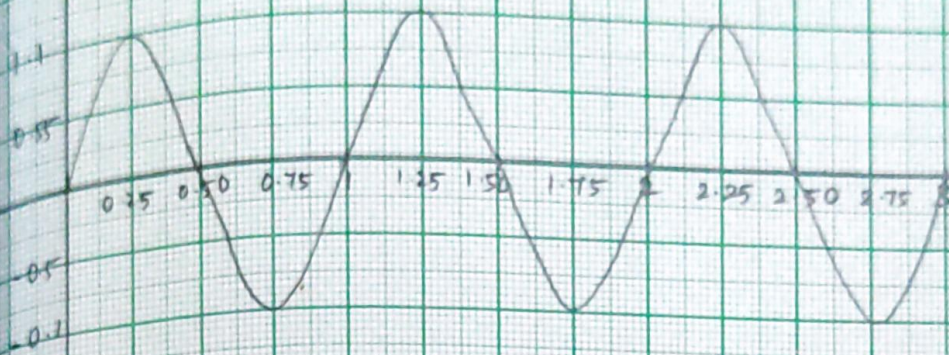
GM

# INPUT POSITIVE CLIPPER

Scale

$$2 \text{ units } 1 \text{ cm} = 0.25 \text{ V}$$

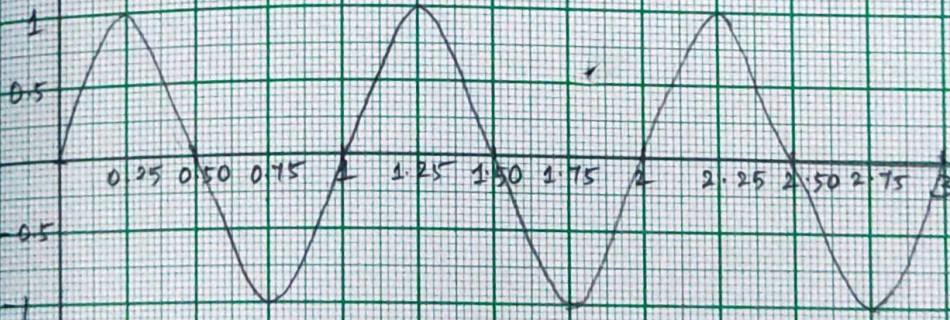
$$17 \text{ units } 1 \text{ cm} = 0.5 \text{ V}$$



## OUTPUT POSITIVE CLIPPER



## INPUT NEGATIVE CLIPPER



## OUTPUT NEGATIVE CLIPPER

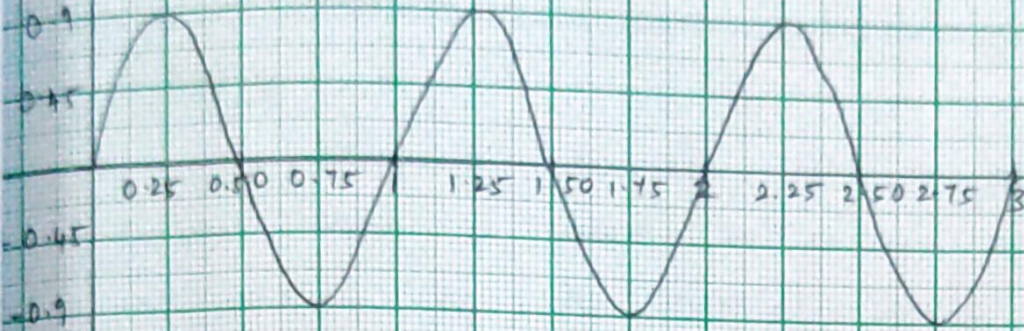


# INPUT SEMI POSITIVE CLIPPER

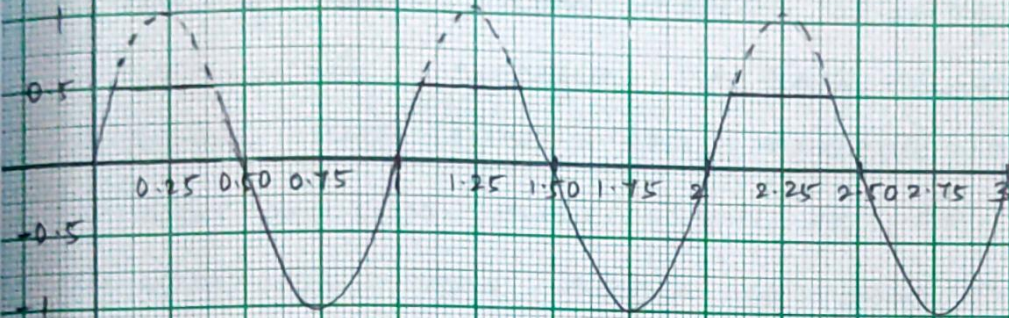
Scale:

1 cm = 0.2

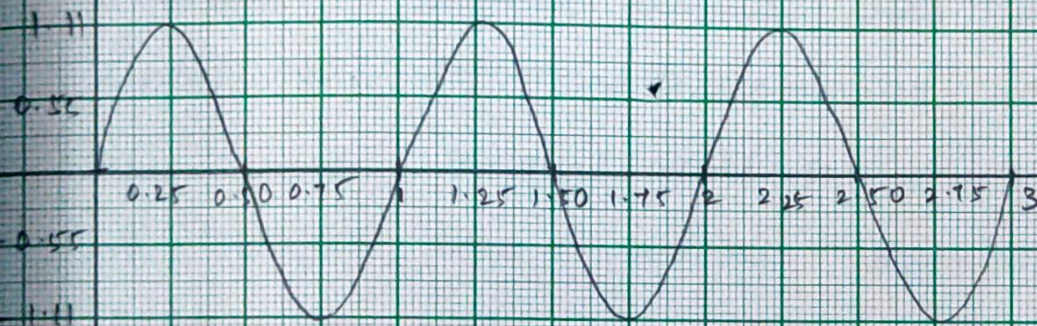
1 cm = 0.1



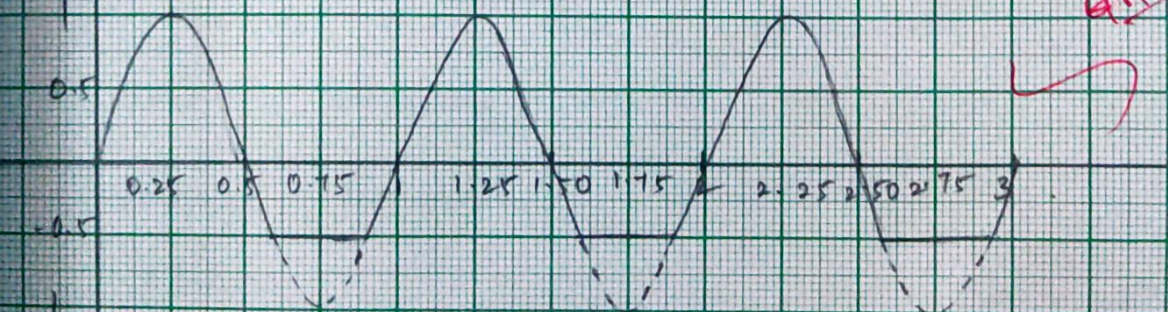
## OUTPUT SEMI POSITIVE CLIPPER



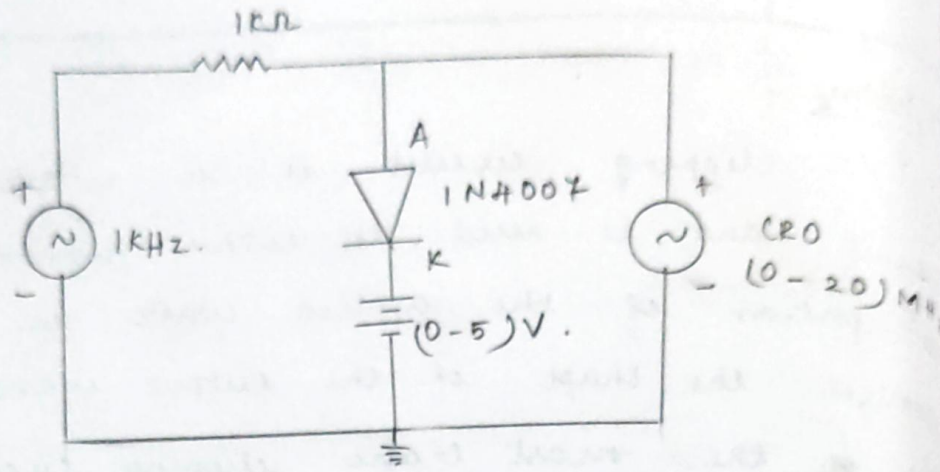
## INPUT SEMI NEGATIVE CLIPPER



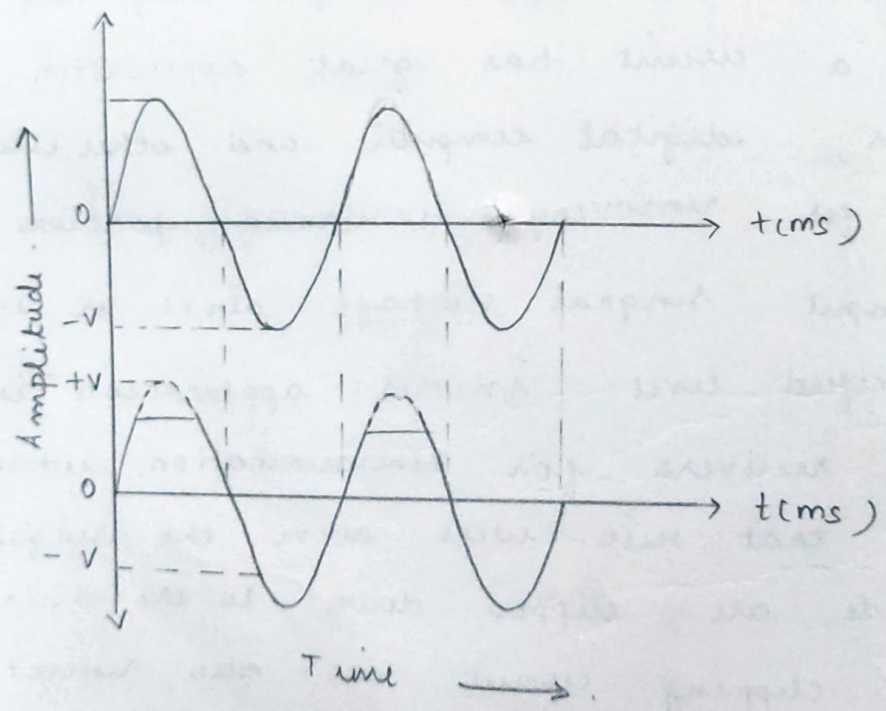
## OUTPUT SEMI NEGATIVE CLIPPER



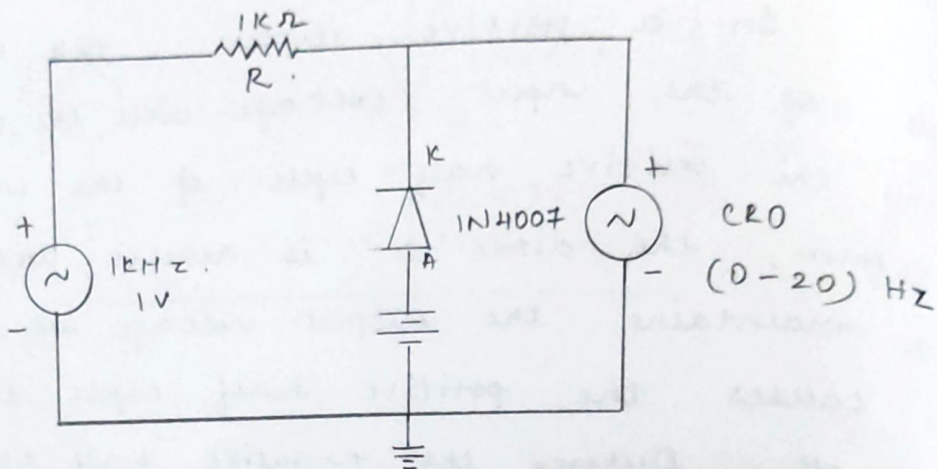
# Positive biased clipper :



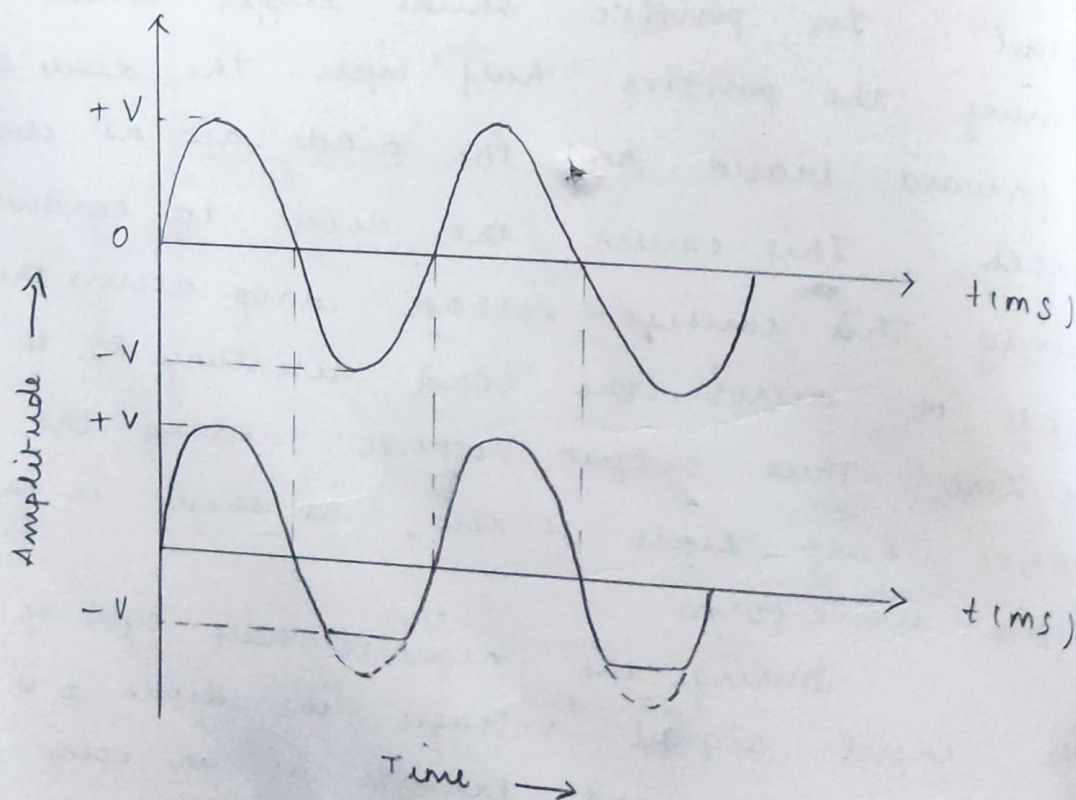
## Model graph :



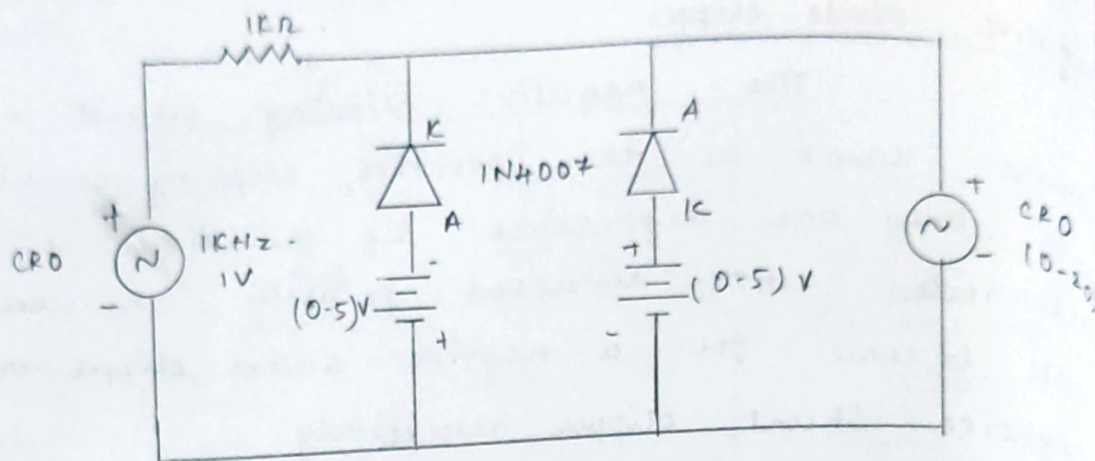
## Negative Biased clipper.



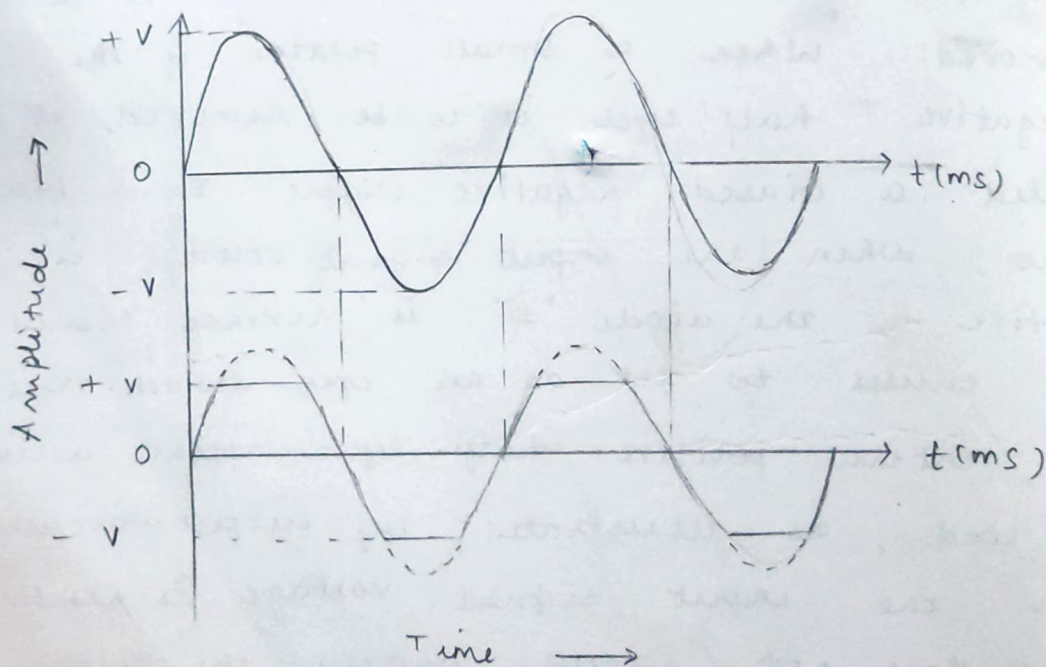
Model graph :



## combinational Biased circuit:-



Model Graph:



# TABULATION:

|                         | INPUT                        |                                         | OUTPUT                       |                                         |
|-------------------------|------------------------------|-----------------------------------------|------------------------------|-----------------------------------------|
|                         | AMPLITUDE                    | TIME                                    | AMPLITUDE                    | TIME                                    |
| POSITIVE CLIPPER        | $2.2 \times 0.5$<br>$= 1.1V$ | $2 \times 0.5$<br>$= 1 \text{ sec}$     | $2.0 \times 0.5$<br>$= 1V$   | $2.0 \times 0.5$<br>$= 1 \text{ sec}$   |
| NEGATIVE CLIPPER        | $2 \times 0.5$<br>$= 1V$     | $2 \times 0.5$<br>$= 1 \text{ sec}$     | $1.8 \times 0.5$<br>$= 0.9V$ | $2 \times 0.5$<br>$= 1 \text{ sec}$     |
| POSITIVE BIAND CLIPPER  | $1.8 \times 0.5$<br>$= 0.9V$ | $2 \times 0.5$<br>$= 1 \text{ sec}$     | $2.0 \times 0.5$<br>$= 1V$   | $2.0 \times 0.5$<br>$= 1 \text{ sec}$   |
| NEGATIVE BIAND CLIPPER  | $2.2 \times 0.5$<br>$= 1.1V$ | $2 \times 0.5$<br>$= 1 \text{ sec}$     | $2 \times 0.5$<br>$= 1V$     | $2 \times 0.5$<br>$= 1 \text{ sec}$     |
| COMBINATIO-NAL CLIPPER. | $2 \times 0.5$<br>$= 1V$     | $1.8 \times 0.5$<br>$= 0.9 \text{ sec}$ | $1.2 \times 0.5$<br>$= 0.6V$ | $1.8 \times 0.5$<br>$= 0.9 \text{ sec}$ |



### Procedure:

- ⇒ Connections are made as per the circuit diagram.
- ⇒ The input waveform is provided from the function generator.
- ⇒ The output waveform is observed in the CRO.
- ⇒ The reading of amplitude and time period for input and output waveform are tabulated.
- ⇒ Graph is plotted between amplitude and time period.
- ⇒ The above procedure is repeated for the other clipper circuits.

### Result:

Thus the positive, negative, biased and combination clippers are constructed using diodes and the output waveform are obtained.

